Unit 1: ER Model

An entity-relationship (ER) model describes data as entities, attributes and relationships. An entity is a real-world object such as a Student or a Course. Attributes describe an entity, for example Name or RollNo. A relationship connects entities, for example Student enrolls in Course. Cardinality defines how many instances can take part: one-to-one, one-to-many or many-to-many. A primary key uniquely identifies each entity instance.

Unit 2: Functional Dependencies, 1NF and 2NF

A functional dependency X -> Y means that the value of X determines the value of Y. Normalization removes redundancy by decomposing tables. First Normal Form (1NF) requires atomic values: no repeating groups or multi-valued cells. Second Normal Form (2NF) requires 1NF and that no non-key attribute depends on only part of a composite primary key (no partial dependency).

Unit 2: Third Normal Form (3NF) and BCNF

Third Normal Form (3NF) requires 2NF and no transitive dependency: a non-key attribute must not depend on another non-key attribute. Example: Student(StudentID, Name, DeptID, DeptName). Here StudentID -> DeptID and DeptID -> DeptName, so DeptName depends transitively on the key. Decompose into Student(StudentID, Name, DeptID) and Department(DeptID, DeptName). BCNF is stricter: for every functional dependency X -> Y, X must be a superkey.

Unit 3: Transactions and ACID

A transaction is a logical unit of work made of one or more SQL operations. ACID properties: Atomicity means all operations happen or none do. Consistency means a transaction moves the database from one valid state to another. Isolation means concurrent transactions do not interfere with each other. Durability means committed changes survive a crash. Commit makes changes permanent; rollback undoes them.

Unit 4: Indexing and B+ Trees

An index is a data structure that speeds up lookups on a column, like the index of a book. A B+ tree keeps keys sorted in a balanced tree whose leaf nodes are linked, which makes range queries efficient. Search, insert and delete take O(log n) time. The trade-off is that indexes use extra storage and slow down writes.